Dataset	$\# F_{total}$	min(|Var(F)|)	max(|Var(F)|)	min(|F|)	max(|F|)
Global	4550	0	8286433	0	7689680
$\hookrightarrow$ Lagniez [?]	1948	5	229100	10	399794
$\hookrightarrow$ Soos [?]	1823	2	8286433	1	7689680
$\hookrightarrow$ Plazar [?]	501	14	486193	31	2598178
$\hookrightarrow$ Sundermann [?]	278	0	31012	0	350120

Table 1: Dataset summary. The first column indicates the dataset, and the $\# F_{total}$ column indicates how many satisfiable formulae the dataset contains. The following columns indicate the minimum and maximum number of variables (resp. clauses) in the dataset.

Dataset	$\# F_{total}$	# D4	# ¬D4	# CapKC ∧ ¬D4	# DivKC ∧ ¬CapKC ∧ ¬D4
Global	4550	185	670	80	55
$\hookrightarrow$ Lagniez [?]	1948	53	213	40	23
$\hookrightarrow$ Soos [?]	1823	100	393	40	28
$\hookrightarrow$ Plazar [?]	501	31	61	0	4
$\hookrightarrow$ Sundermann [?]	278	1	3	0	0

Table 2: Experimental results regarding the scalability of DivKC and CapKC. Column $\# F_{total}$ indicates the total number of formulae in each dataset. The next column shows the number of formulae compiled only by D4 [?] but not by DivKC or CapKC. Column # ¬D4 shows the number of formulae not compiled by D4. Column # CK ∧ ¬D4 shows the number of formulae that are compiled by CapKC but not by D4. The last column indicates the number of formulae that were only compiled by DivKC, but not by D4 or CapKC.

Dataset	$\# F_{total}$	# ¬D4	# ¬D4 ∧ CapKC	# ¬D4 ∧ DivKC	# CapKC ∧ ¬DivKC ∧ ¬D4
Global	4550	670	80	114	21
$\hookrightarrow$ Lagniez [?]	1948	213	40	52	11
$\hookrightarrow$ Soos [?]	1823	393	40	58	10
$\hookrightarrow$ Plazar [?]	501	61	0	4	0
$\hookrightarrow$ Sundermann [?]	278	3	0	0	0

Table 3: Experimental results regarding the scalability of DivKC and CapKC.

Dataset	# F	$Y_l \leq \|R_F\|$	$Y_h \geq \|R_F\|$	Coverage	$\|R_{G_P}\| \leq \|R_F\| \leq \|R_{G_U}\|$
Global	3670	0.969	0.857	0.826	1.000
$\hookrightarrow$ Lagniez [?]	1689	0.969	0.895	0.864	1.000
$\hookrightarrow$ Soos [?]	1307	0.969	0.906	0.875	1.000
$\hookrightarrow$ Plazar [?]	403	0.968	0.789	0.757	1.000
$\hookrightarrow$ Sundermann [?]	271	0.967	0.487	0.454	1.000

Table 4: Experimental results for DivKC-AMC. Column # F indicates with how many formulae the statistics have been computed. The 'Coverage' column indicates how often $|R_F|$ is within the confidence interval $[Y_l; Y_h]$ and thus measures the accuracy of our method.

Dataset	# F	$Y_l \leq \|R_F\|$	$Y_h \geq \|R_F\|$	Coverage	$\|R_F\| \leq \|R_{G_U}\|$
Global	323	0.994	0.988	0.981	1.000
$\hookrightarrow$ Lagniez [?]	168	0.994	0.988	0.982	1.000
$\hookrightarrow$ Soos [?]	149	1.000	0.987	0.987	1.000
$\hookrightarrow$ Plazar [?]	4	1.000	1.000	1.000	1.000
$\hookrightarrow$ Sundermann [?]	2	0.500	1.000	0.500	1.000

Table 5: Experimental results for CapKC. Column # F indicates with how many formulae the statistics have been computed. The 'Coverage' column indicates how often $|R_F|$ is within the confidence interval $[Y_l; Y_h]$ and thus measures the accuracy of our method.

Dataset	# F	$Y_l \geq \|R_{G_P}\|$	$Y_h \leq \|R_{G_U}\|$	Both	median (r_c)	max (r_c)
Global	3669	0.807	0.836	0.721	4.17e-9	0.54
↪ Lagniez [?]	1689	0.844	0.869	0.760	1.8e-9	0.54
↪ Soos [?]	1307	0.920	0.893	0.834	1.75e-7	0.0962
↪ Plazar [?]	403	0.697	0.769	0.610	1.64e-13	0.0153
↪ Sundermann [?]	270	0.196	0.448	0.093	1.69e-13	0.281

Table 6: Experimental results comparing the bounds obtained with DivKC-AMC and with Lemma ??. Column # F indicates with how many formulae the statistics have been computed. The 'Both' column indicates how often we have $Y_l \geq |R_{G_P}| \wedge Y_l \leq |R_F|$ and $Y_h \leq |R_{G_U}| \wedge Y_h \geq |R_F|$. The last two columns represent the observed median and maximum values of the ratio $r_c = \frac{min(Y_h, |R_{G_U}|) - max(Y_l, |R_{G_P}|)}{|R_{G_U}| - |R_{G_P}|}$, which was calculated exclusively if $Y_l \leq |R_F| \leq Y_h$. The number of formulae on which the last two columns are computed can easily be obtained by multiplying the # F column with the 'Coverage' column in Table 4.

Dataset	# F	$Y_h \leq \|R_{G_U}\|$	median (r_c)	max (r_c)
Global	334	0.994	0.00794	0.0257
↪ Lagniez [?]	175	0.994	0.0072	0.0257
↪ Soos [?]	153	0.993	0.00837	0.0257
↪ Plazar [?]	4	1.000	0.00684	0.0257
↪ Sundermann [?]	2	1.000	0.00981	0.0155

Table 7: Experimental results comparing the bounds obtained with CapKC. Column # F indicates with how many formulae the statistics have been computed. The last two columns represent the observed median and maximum values of the relative confidence-interval width $r_c = \dfrac{Y_h - Y_l}{|R_{G_U}|}$.

Dataset	# F	$l \leq Y_{ApproxMC} \leq h$	$l \leq Y_{DivKC\text{-}AMC} \leq h$	$l \leq Y_{CapKC\text{-}AMC} \leq h$
Global	2814	0.939	0.875	0.982
↪ Lagniez [?]	1353	0.982	0.891	0.981
↪ Soos [?]	1186	0.957	0.848	0.979
↪ Plazar [?]	253	0.613	0.905	0.996
↪ Sundermann [?]	22	1.000	0.955	1.000

Table 8: Experimental results comparing the accuracy of DivKC-AMC and CapKC-AMC with ApproxMC 7. Column # F indicates with how many formulae the statistics have been computed. Column $l \leq Y_A \leq h$ indicates how often the model count returned by A is within the indicated bounds, with $l = \dfrac{|R_F|}{1.2}$ and $h = 1.2 \times |R_F|$.

Dataset	# F	$l \leq Y_{ApproxMC} \leq h$	$l \leq Y_{DivKC\text{-}AMC} \leq h$	$l \leq Y_{CapKC\text{-}AMC} \leq h$
Global	298	0.997	0.500	0.826
↪ Lagniez [?]	148	1.000	0.486	0.824
↪ Soos [?]	147	0.993	0.503	0.830
↪ Plazar [?]	3	1.000	1.000	0.667
↪ Sundermann [?]	0			

Table 9: Experimental results comparing the accuracy of DivKC-AMC and CapKC-AMC with ApproxMC 7. Column # F indicates with how many formulae the statistics have been computed. Column $l \leq Y_A \leq h$ indicates how often the model count returned by A is within the indicated bounds, with $l = \dfrac{|R_F|}{1.2}$ and $h = 1.2 \times |R_F|$. We only consider formulae for which $|R_{G_U}| \neq |R_F|$, thereby excluding cases where $CapKC$ returns $G_U = F$ and the model count can be obtained directly from G_U.

Dataset	# F	# DivKC-AMC	$\log_{10}$(min)	mean	median	$\log_{10}$(max)
Global	3291	2122	-5.4	153.0	1.8	5.1
↪ Lagniez [?]	1623	1156	-2.9	224.2	2.1	4.9
↪ Soos [?]	1358	851	-3.5	6.2	1.7	3.6
↪ Plazar [?]	288	98	-5.4	1.0	0.3	1.2
↪ Sundermann [?]	22	17	-1.2	5954.6	12.6	5.1

Table 10: Experimental results comparing the runtime of DivKC-AMC with ApproxMC 7 for 20 runs. Column # F indicates the number of formulae over which the statistics were computed, and column # DivKC-AMC shows how often DivKC-AMC was faster than ApproxMC 7. The remaining columns report how much faster DivKC-AMC was, based on the logarithm of the minimum, the mean, the median, and the logarithm of the maximum of the ratio: ApproxMC 7 execution time divided by DivKC-AMC execution time.

Dataset	# F	# CapKC-AMC	$\log_{10}$(min)	mean	median	$\log_{10}$(max)
Global	2838	1629	-3.5	162.7	1.6	5.0
↪ Lagniez [?]	1352	820	-2.6	251.6	1.8	4.7
↪ Soos [?]	1204	679	-2.1	6.1	1.5	3.2
↪ Plazar [?]	259	111	-3.5	1.5	0.7	1.2
↪ Sundermann [?]	23	19	-1.2	4948.0	10.1	5.0

Table 11: Experimental results comparing the runtime of CapKC-AMC with ApproxMC 7 for 20 runs. Column # F indicates the number of formulae over which the statistics were computed, and column # CapKC-AMC shows how often CapKC-AMC was faster than ApproxMC 7. The remaining columns report how much faster CapKC-AMC was, based on the logarithm of the minimum, the mean, the median, and the logarithm of the maximum of the ratio: ApproxMC 7 execution time divided by CapKC-AMC execution time.

Dataset	# F	# CapKC-AMC	$\log_{10}$(min)	mean	median	$\log_{10}$(max)
Global	294	2	-2.1	1.9	0.0	2.7
↪ Lagniez [?]	143	2	-2.1	3.8	0.0	2.7
↪ Soos [?]	149	0	-2.1	0.2	0.0	-0.1
↪ Plazar [?]	1	0	-1.2	0.1	0.1	-1.2
↪ Sundermann [?]	1	0	-0.3	0.5	0.5	-0.3

Table 12: CapKC vs Approxmc7.